



1. What is the value of $2012^2 - 2011^2 + 2010^2 - 2009^2 + \dots + 2^2 - 1^2$?

- (A) 2000000 (B) 2023066 (C) 2024072 (D) 2025078 (E) 2027091

2. When the decimal point of a certain positive number is moved four places to the right, the new number is nine times the reciprocal of the original number. What was the original number?

- (A) 0.0003 (B) 0.003 (C) 0.03 (D) 0.3 (E) 3

3. For real numbers x and y , define $x \oplus y = \frac{3x+5y}{4+3xy}$. What is the value of a such that $4 \oplus a = \frac{1}{2}$?

- (A) 10 (B) 12 (C) 14 (D) 16 (E) 18

4. How many three-digit positive integers are there such that the product of their digits is a prime number?

- (A) 6 (B) 8 (C) 10 (D) 11 (E) 12

5. Let $ABCDE$ be a regular pentagon. Segments BD and CE intersect at F . What is the size of angle $\angle DFC$?

- (A) 90° (B) 96° (C) 100° (D) 108° (E) 120°

6. Let P be a polynomial such that $P(x+3) = 3x^2 + 7x + 4$. If $P(x) = ax^2 + bx + c$, what is the value of $a - b + c$?

(A) 2 (B) 4 (C) 10 (D) 20 (E) 24

7. Let $ABCD$ be a square. Let M be the midpoint of side AB , and E be the intersection of segments CM and BD . What is the ratio $\frac{EB}{EM}$?

(A) $\sqrt{\frac{3}{2}}$ (B) $\sqrt{\frac{8}{5}}$ (C) $\sqrt{\frac{5}{3}}$ (D) $\sqrt{\frac{7}{4}}$ (E) 2

8. Let P be a polynomial such that $P(x) = ax^4 - bx^2 + x + 5$, where a and b are constants. If $f(-3) = 2$, what is the value of $f(3)$?

(A) 3 (B) 4 (C) 6 (D) $\frac{13}{2}$ (E) 8

9. Let ABC be a triangle with $\angle ABC = 90^\circ$. Let D be a point on AC such that BD is perpendicular to AC . If $AD = 3$ and $CD = 4$, what is the area of triangle ABC ?

(A) 11 (B) $8\sqrt{2}$ (C) $7\sqrt{3}$ (D) $\frac{25}{2}$ (E) 13

10. Let a and b be the two roots of the equation $x^2 - 7x + 10 = 0$. What is the value of $a^5 + b^5$?

(A) 1357 (B) 1573 (C) 3157 (D) 3571 (E) 5371

11. Let a and b be the roots of the equation $x^2 - mx + 2 = 0$. If $a + \frac{1}{b}$ and $b + \frac{1}{a}$ are the roots of the equation $x^2 - px + q = 0$, what is the value of q ?

(A) 2 (B) $\frac{5}{2}$ (C) 4 (D) $\frac{9}{2}$ (E) 6

12. Consider the sequence $1, 2, 1, 2, 2, 1, 2, 2, 2, 1, 2, 2, 2, 2, 1, \dots$. What is the sum of the first 1234 terms of it?

(A) 2397 (B) 2419 (C) 2425 (D) 2440 (E) 2452

13. Let x and y be two distinct real numbers such that $\frac{x}{y} + \frac{x+10y}{y+10x} = 2$. What is the sum of all possible values of $\frac{x}{y}$?

(A) 0.8 (B) 1.0 (C) 1.4 (D) 1.8 (E) 2.0

14. Let ABC and DEF be equilateral triangles. D lies on segment BC , E lies on segment CA , and F lies on segment AB . Suppose that DE is perpendicular to BC . If x is the area of triangle ABC , and y is the area of triangle DEF , what is the value of $\frac{x}{y}$?

(A) $\frac{5}{2}$ (B) $2\sqrt{2}$ (C) 3 (D) $2\sqrt{3}$ (E) $3\sqrt{2}$

15. Among the numbers $1!, 1! + 2!, 1! + 2! + 3!, \dots, 1! + 2! + 3! + \dots + 2012!$, how many are perfect squares?

(A) 2 (B) 4 (C) 502 (D) 503 (E) 1006

16. Let $a_2, a_3, a_4, a_5, a_6, a_7$ be integers such that $\frac{5}{7} = \frac{a_2}{2!} + \frac{a_3}{3!} + \frac{a_4}{4!} + \frac{a_5}{5!} + \frac{a_6}{6!} + \frac{a_7}{7!}$, with $0 \leq a_i < i$ for $i = 2, 3, \dots, 7$. What is the value of $\sum_{i=2}^7 a_i^2$?

(A) 6 (B) 12 (C) 18 (D) 23 (E) 27

17. For a positive integer n , let $E(n)$ be the sum of the digits of n which are even. For example, $E(21483) = 2 + 4 + 8 = 14$. What is the value of $\sum_{i=1}^{100} E(i)$?

(A) 400 (B) 450 (C) 500 (D) 550 (E) 600

18. A sequence of numbers $t_1, t_2, t_3, \dots$ is defined by $t_1 = 1$, and $t_n = \frac{t_{n-1}+2}{t_{n-1}+1}$ for $n = 2, 3, \dots$. As n gets larger and larger, what value does t_n approach?

(A) $\sqrt{2}$ (B) $\frac{13}{2+\sqrt{5}}$ (C) $\frac{14}{3+2\sqrt{2}}$ (D) $\frac{3}{2}$ (E) $\frac{4}{1+\sqrt{3}}$

19. How many ordered 4-tuples of positive integers (a, b, c, d) are there such that $a, b, c, d \leq 12$ and $a + 2b + 3c + 4d$ is divisible by 12?

(A) 320 (B) 648 (C) 960 (D) 1320 (E) 1728

20. Let ABC be a triangle such that $AB = 9$ and $AC = 7$. Let D be the midpoint of segment BC . Let X be a point on segment AB such that $DX \perp AB$. Let Y be a point on segment AC such that $DY \perp AC$. If $AX = 6$, what is the length of AY ?

(A) $\frac{30}{7}$ (B) $\frac{32}{7}$ (C) $\frac{34}{7}$ (D) $\frac{36}{7}$ (E) $\frac{38}{7}$

21. The line $7y - x = 5$ is tangent to the circle $x^2 + y^2 - 5x + 5y = 0$. Another line, $y = mx + c$, is also tangent to the same circle, and parallel to the given line. What is the value of $m - c$?

(A) $\frac{46}{7}$ (B) 7 (C) $\frac{43}{6}$ (D) $\frac{22}{3}$ (E) $\frac{15}{2}$

22. Let $ABCD$ be a quadrilateral such that $AB = \sqrt{5}$, $BC = \sqrt{13}$, $CD = \sqrt{17}$, $DA = \sqrt{29}$, and $AC = \sqrt{10}$. What is the area of the quadrilateral $ABCD$?

(A) 9 (B) $\sqrt{85}$ (C) 10 (D) $6\sqrt{2}$ (E) $\sqrt{145}$

23. Let $a > b > c > d > 0$ be real numbers. Suppose that $\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} = \frac{13}{2}$ and $\frac{a}{c} + \frac{c}{a} + \frac{b}{d} + \frac{d}{b} = 9$. What is the value of $\frac{a}{b} + \frac{c}{d}$?

(A) 4 (B) 4.5 (C) 5 (D) 5.5 (E) 6

24. For any positive integer n , let $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$. Suppose that $7.4854 < H_{1000} < 7.4855$. What is the greatest positive integer less than $H_1 + H_2 + \dots + H_{1000}$?

(A) 6484 (B) 6486 (C) 6488 (D) 6490 (E) 6492

25. The smallest possible distance between a point P on the line $y = x$ and a point Q on the curve $y = 2^x$ is in which of the following range?

(A) $[0.5, 0.6)$ (B) $[0.6, 0.7)$ (C) $[0.7, 0.8)$ (D) $[0.8, 0.9)$ (E) $[0.9, 1.0)$

Time limit: 75 minutes